**Data Structures**

BS (CS) _Fall_2024

# Lab_07 Tasks




## Learning Objectives:

1. Doubly Linked List
2. Circular Linked List

# Lab Tasks

## Question 1

Design a doubly linked list class with the following specifications:

1. DoublyNode Class:

    - Create a class called `DoublyNode` with the following data members:

        - int data to store the data for the node.
        - DoublyNode* prev to store a pointer to the previous node.
        - DoublyNode* next to store a pointer to the next node.

    - Provide the following member functions:

        - A default constructor for `DoublyNode.
        - A parameterized constructor that sets the data, prev, and next pointers.
        - Getter methods for data, prev, and next.
        - Setter methods for data, prev, and next.

2. DoublyLinkedList Class:

    - Create a class called `DoublyLinkedList` to manage a doubly linked list. It should have a pointer to the head node initially.

    - Provide the following member functions:

        - A default constructor for DoublyLinkedList.

        - A function called `insert` to insert a data item at the end of the doubly linked list. If the list is empty, the new node becomes the head.

- A function called insertToHead to insert a data element at the beginning of the doubly linked list.
- A function called isEmpty to check if the doubly linked list is empty.
- Explain why there is no need to implement an `isFull` function.
- A function called `search` to search for a specific element provided by the user and return true if found, false otherwise.
- A function called `update` to update/replace one data element with another.
- A function called `insertAtIndex` that will count the data elements in the doubly linked list and place a new data element in the specified location (indexed by counting from 0).
- A function called `deleteData` to delete a specified data element. This should involve updating the previous node's `next` pointer and the next node's `prev` pointer accordingly.
- A function called `print` to print the data of the doubly linked list from the head to the end.

Your task is to implement both the `DoublyNode` and `DoublyLinkedList` classes based on the provided specifications and ensure that they work correctly as described.

# Question 2

Design a circular linked list class with the following specifications:

1. Node Class:
    - Create a class called `Node` with the following data members:
        - int data to store the data for the node.
        - `Node* next` to store a pointer to the next node.
    - Provide the following member functions:
        - A default constructor for `Node`.
        - A parameterized constructor that sets the data and next pointer.
        - Getter methods for data and next.

    - Setter methods for data and next.

2. CircularLinkedList Class:

    - Create a class called `CircularLinkedList` to manage a circular linked list. It should have a pointer to the head node initially.

    - Provide the following member functions:

        - A default constructor for `CircularLinkedList`.

        - A function called `insert` to insert a data item at the end of the circular linked list. If the list is empty, the new node becomes the head, and it should point to itself.

        - A function called `isEmpty` to check if the circular linked list is empty.

        - A function called `search` to search for a specific element provided by the user and return true if found, false otherwise.

        - A function called `update` to update/replace one data element with another.

        - A function called `insertAtIndex` that will count the data elements in the circular linked list and place a new data element in the specified location (indexed by counting from 0).

        - A function called `deleteData` to delete a specified data element.

        - A function called `print` to print the data of the circular linked list starting from the head node and continuing until it reaches the head node again.

Your task is to implement both the `Node` and `CircularLinkedList` classes based on the provided specifications and ensure that they work correctly as described.